

Problem Set 1 - Solutions

Knowledge List: 4, 5, 6, 12

Beliefs List: 1, 2, 3, 8

ECN 403/503

Fall 2022



Knowledge Problems:

Problem 4

$$(a) \quad \mathcal{P}_1 = \{ \{1\}, \{2,3\}, \{4,5\}, \{6,7\} \} \quad \mathcal{P}_2 = \{ \{1,2\}, \{3,4\}, \{5,6\}, \{7\} \}$$

$$T_1 = \begin{matrix} \updownarrow & \updownarrow & \updownarrow & \updownarrow \\ a & b & c & d \end{matrix} \quad T_2 = \begin{matrix} \updownarrow & \updownarrow & \updownarrow & \updownarrow \\ \alpha & \beta & \delta & \delta \end{matrix}$$

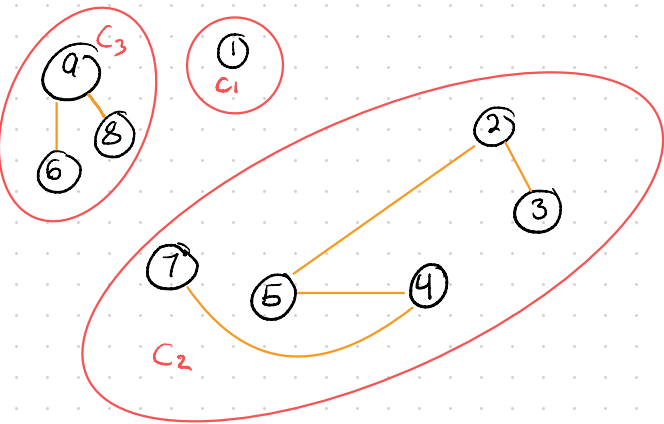
Arrows indicate the respective maps τ_i

$$(b) \quad K_1 A = \{1\} = A, \quad K_2 A = \emptyset; \quad K_1 B = \{1,2,3,4,5\}, \quad K_2 B = \{1,2,3,4,5,6\} = B$$

$$K_2 K_1 A = K_2 A = \emptyset, \quad K_1 K_2 A = \emptyset; \quad K_2 K_1 B = K_2 \{1,2,3,4,5\} = \{1,2,3,4\}, \quad K_1 K_2 B = K_1 B = \{1,2,3,4,5\}$$

$$K_1 K_2 K_1 B = K_1 \{1,2,3,4\} = \{1,2,3\}, \quad K_2 K_1 K_2 B = K_2 \{1,2,3,4,5\} = \{1,2,3,4\}$$

Problem 5 The graph G_C



The common knowledge algebra: $\Sigma_C = \{\emptyset, \Omega, C_1, C_2, C_3, C_1 \cup C_2, C_1 \cup C_3, C_2 \cup C_3\}$

• At ω_1 the events that are common knowledge are:

$\{A \in 2^\Omega : C_1 \subseteq A\}$ i.e. all events A that contain the connected component containing $\omega = 1$

• At $\omega = 9$, the events that are common knowledge are $\{A \in 2^\Omega : C_3 \subseteq A\}$

• At $\omega = 5$, the events that are common knowledge are $\{A \in 2^\Omega : C_2 \subseteq A\}$

Problem 6

Consider the situation in problem 2 and the event "Juliet read the letter".

Problem 12

According to Thm 1.21 A is common knowledge for every state of the world in A if any of the following three equivalent statements are true:

- (1) For every $w \in A$, and for any sequence $i_1, i_2, \dots, i_k \in N$ it holds $w \in K_{i_1} \dots K_{i_k} A$,
that is: $K_{i_1} \dots K_{i_k} A = A$
- (2) For every $w \in A$, $C(w) \subseteq A$, or $\bigcup_{w \in A} C(w) \subseteq A$
- (3) For every $w \in A$, $w \in K_c A$ or $K_c A = A$

Now, consider statement (4) below:

- (4) For every $w \in A$, $w \in K_1 K_2 \dots K_n A$ or $K_1 K_2 \dots K_n A = A$

We need to show that (4) is equivalent to any of (1), (2), or (3) above.

First we show that (1) $\Rightarrow$ (4). Let $k=n$ and let the sequence $i_1, \dots, i_n$ be $1, 2, \dots, n$. Then (1) implies $K_1 K_2 \dots K_n A = A$ which is what we wanted to show.

Now we show that (4) $\Rightarrow$ (3).

Let $K_1 K_2 \dots K_n A = A$. $\Rightarrow K_i A = A$ for all $i = 1, \dots, n$, otherwise we would have had $K_1 \dots K_n A \subsetneq A$

$K_i A = A$ for all $i \in N$ implies that $A \in \Sigma_c \Rightarrow K_c A = A$. Since we saw that (3) $\Rightarrow$ (1) in the proof of Thm 1.20 (Problem 11), we are done.

Belief Problems

Problem 1 First note that $\Sigma_c = \{\emptyset, \Omega\}$

P1 Information	Posteriors $\mu_1(\omega \Sigma_1)$			
	ω_1	ω_2	ω_3	ω_4
$\{\omega_1, \omega_2\}$	$\frac{1}{2}$	$\frac{1}{2}$	0	0
$\{\omega_3, \omega_4\}$	0	0	$\frac{1}{2}$	$\frac{1}{2}$

P2 information	Posteriors $\mu_2(\omega \Sigma_2)$			
	ω_1	ω_2	ω_3	ω_4
$\{\omega_1, \omega_2, \omega_3\}$	$\frac{1}{3}$	$\frac{1}{3}$	$\frac{1}{3}$	0
$\{\omega_4\}$	0	0	0	1

Let $A = \{\omega_1, \omega_3\}$.

Then (a) $\mu_1(A | \Sigma_1)(\omega) = \frac{1}{2}$ for all $\omega \in \Omega$

$$(b) \mu_2(A | \Sigma_2)(\omega) = \begin{cases} \frac{2}{3} & \text{if } \omega = \{\omega_1, \omega_2, \omega_3\} \\ 0 & \text{if } \omega = \omega_4 \end{cases}$$

Since $\{\omega \in \Omega : \mu_1(A | \Sigma_1)(\omega) = \frac{1}{2}\} = \Omega \in \Sigma_c$

the fact that P1 ascribes prob. $\frac{1}{2}$ to A
is common knowledge.

Note that $\{\omega \in \Omega : \mu_2(A | \Sigma_2)(\omega) = \frac{2}{3}\} = \{\omega_1, \omega_2, \omega_3\}$
which is not common knowledge.

(c) The example doesn't contradict Aumann's Theorem because although there is a common prior, the posteriors are not common knowledge.

Problem 2 Again, note that $\Sigma_c = \{\emptyset, \Omega\}$

P1 Information	Posteriors $\mu_1(\omega \Sigma_1)$			
	ω_1	ω_2	ω_3	ω_4
$\{\omega_1, \omega_2\}$	$1/2$	$1/2$	0	0
$\{\omega_3, \omega_4\}$	0	0	$1/2$	$1/2$

Let $A = \{\omega_1, \omega_3\}$

$$\mu_1(A | \Sigma_1)(\omega) = \frac{1}{2} \text{ for all } \omega \in \Omega$$

$$\text{Notice } \{\omega \in \Omega : \mu_1(A | \Sigma_1)(\omega) = \frac{1}{2}\} = \Omega$$

Thus, it is common knowledge that both players have different posteriors about A , so they "agree to disagree".

This doesn't contradict Aumann's Theorem because the priors were not common.

P2 information	Posteriors $\mu_2(\omega \Sigma_2)$			
	ω_1	ω_2	ω_3	ω_4
$\{\omega_1, \omega_4\}$	$2/5$	0	0	$3/5$
$\{\omega_2, \omega_3\}$	0	$3/5$	$2/5$	0

$$\mu_2(A | \Sigma_2)(\omega) = 2/5 \text{ for all } \omega \in \Omega$$

$$\text{Notice } \{\omega \in \Omega : \mu_2(A | \Sigma_2)(\omega) = 2/5\} = \Omega$$

that both players have different posteriors

Problem 3

P1 Information	Posteriors $\mu_1(\omega \Sigma_1)$			
	ω_1	ω_2	ω_3	ω_4
$\{\omega_1, \omega_2\}$	$2/3$	$1/3$	0	0
$\{\omega_3, \omega_4\}$	0	0	$1/3$	$2/3$

P2 information	Posteriors $\mu_2(\omega \Sigma_2)$			
	ω_1	ω_2	ω_3	ω_4
$\{\omega_1, \omega_2, \omega_3\}$	$1/2$	$1/4$	$1/4$	0
$\{\omega_4\}$	0	0	0	1

Let $A = \{\omega_2, \omega_3\}$

$$\mu_1(A | \Sigma_1)(\omega) = \frac{1}{3} \text{ at all } \omega \in \Omega \quad \mu_2(A | \Sigma_2)(\omega) = \begin{cases} \frac{1}{2} & \text{at } \omega \in \{\omega_1, \omega_2, \omega_3\} \\ 0 & \text{at } \omega = \omega_4 \end{cases}$$

Thus, at ω_3 $q_1^0 = \frac{1}{3}$ and $q_2^0 = \frac{1}{2}$

After announcing q_i^0 :

- P2 knows $\{\omega \in \Omega : q_1^0 = \frac{1}{3}\} = \Omega$ so this adds no new information $\rightarrow q_2^1 = \frac{1}{2}$

- P1 knows $\{\omega \in \Omega : q_2^0 = \frac{1}{2}\} = \{\omega_1, \omega_2, \omega_3\}$ so now P1 deems ω_4 impossible, then he knows, given his information, that the true state is $\omega_3 \rightarrow q_1^1 = 1$

After announcing q_i^1 :

- P2 knows that the states cannot be $\{\omega_1, \omega_2\}$ for otherwise P1 would have announced $q_1^1 = \frac{1}{3}$, thus P2 doesn't deem possible states $\{\omega_1, \omega_2\}$ and only deems possible $\omega_3 \rightarrow q_2^2 = 1$

- P1 puts $q_1^2 = q_1^1 = 1$.

Then, for $i=1, 2, t \geq 2, q_i^t = 1 \rightarrow q_i^t$ converges and they are equal between them.

For the next cases, let's compute the actual random variable representing the posterior:

$$Q_1^0 = \mu(A|Z_1) = \frac{1}{3} \text{ for all } \omega \in \Omega$$

↓

$$Q_1^1 = \mu(A|Z_1, Q_1^0) = \begin{cases} \frac{1}{3} & \text{if } \omega \in \{\omega_1, \omega_2\} \\ 1 & \text{if } \omega = \omega_3 \\ 0 & \text{if } \omega = \omega_4 \end{cases}$$

↓

$$Q_1^2 = \mu(A|Z_1, Q_1^0, Q_1^1) = \begin{cases} \frac{1}{3} & \text{if } \omega \in \{\omega_1, \omega_2\} \\ 1 & \text{if } \omega = \omega_3 \\ 0 & \text{if } \omega = \omega_4 \end{cases}$$

$$Q_2^0 = \mu(A|Z_2) = \begin{cases} \frac{1}{2} & \omega \in \{\omega_1, \omega_2, \omega_3\} \\ 0 & \omega = \omega_4 \end{cases}$$

↓

$$Q_2^1 = \mu(A|Z_2, Q_2^0) = \begin{cases} \frac{1}{2} & \text{if } \omega \in \{\omega_1, \omega_2, \omega_3\} \\ 0 & \text{if } \omega = \omega_4 \end{cases}$$

↓

$$Q_2^2 = \mu(A|Z_2, Q_2^0, Q_2^1) = \begin{cases} \frac{1}{3} & \omega \in \{\omega_1, \omega_2\} \\ 1 & \text{if } \omega = \omega_3 \\ 0 & \text{if } \omega = \omega_4 \end{cases}$$

And the process has stabilized.

Thus at either $\omega = \omega_1, \omega_2$, $(q_1^0, q_2^0) = (1/3, 1/2)$, $(q_1^1, q_2^1) = (1/3, 1/2)$, $(q_1^2, q_2^2) = (1/3, 1/3)$

and $(q_1^t, q_2^t) = (1/3, 1/3)$ for $t \geq 2$

Problem 8

Assume μ is a common prior and $\mu(\omega) > 0 \quad \forall \omega \in \Omega$

We need to show $\mu(A | P_i(\omega)) = 1 \Leftrightarrow P_i(\omega) \subseteq A$

$\Leftarrow$ Direction $P_i(\omega) \subseteq A$, take conditional probabilities with $P_i(\omega)$ as the conditioning event:

$$1 = \mu(P_i(\omega) | P_i(\omega)) \leq \mu(A | P_i(\omega)) \leq 1 \Rightarrow \mu(A | P_i(\omega)) = 1$$

monotonicity of probability

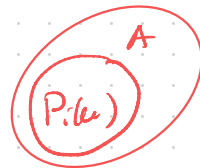
Prob. always less than or equal to 1

$\Rightarrow$ Direction $\mu(A | P_i(\omega)) = 1 \Rightarrow \frac{\mu(A \cap P_i(\omega))}{\mu(P_i(\omega))} = 1 \Rightarrow \mu(A \cap P_i(\omega)) = \mu(P_i(\omega))$

Since $\mu(\omega) > 0$ the last equality implies that $P_i(\omega) \subseteq A$.

why this is relevant?

$\mu(\omega) > 0$ implies that the "situation" looks like this



instead of

